Development of a Deep Learning Application for Weed Detection and Localization for a Small-scale Weeder

by

Mudalpath Mudiyanselage Pasindu Kavinga

Department of Agriculture Engineering

Faculty of Agriculture

University of Peradeniya

Peradeniya

Sri Lanka

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DEDICATION

This thesis is dedicated to my parents, my brother, my friends who supported me in difficult times, and teachers who guided me throughout this journey.

ABSTRACT

Weed management is a critical challenge in modern agriculture, often requiring significant labor and chemical usage that negatively impacts crop yield and the environment. This project presents the development of an automated vision-guided robotic weeding system designed to identify and mechanically remove weeds with minimal human intervention. The system integrates an AI-based weed detection model, trained using a custom dataset and fine-tuned YOLOv8 architecture on Google Colab with an NVIDIA T4 GPU, to achieve accurate weed identification.

The hardware design consists of a three-axis linear actuator mechanism mounted on a four-wheel steel chassis, driven by NEMA 17 stepper motors with A4988 motor drivers controlled via an ESP32 microcontroller. A web camera provides real-time visual feedback for weed detection, while a rack-and-pinion-based gripper performs the physical weed removal. The system's control interface was developed in Python communicating serially with the ESP32 for motion and grip control.

Experimental testing evaluated travel accuracy, movement speed, and weed removal efficiency. Results showed that the system achieved uprooting accuracy of 27.5%, an average operation time of 42.96 seconds per weed, and a detection accuracy exceeding 93% depending on confidence threshold settings. These findings demonstrate the feasibility of combining computer vision and low-cost automation for precision weeding in early-stage maize field.

The proposed system significantly reduces manual effort and chemical usage, contributing to sustainable agricultural practices. Further enhancements, including closed-loop motion feedback, improved gripper design, and adding onboard computing hardware, are recommended to advance the prototype toward a fully autonomous field-ready solution.

Keywords: Robotic weeder, Precision agriculture, Weed detection, Computer vision

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CAGR	Compound Annual Growth Rate
CNN	Convolutional Neural Network
ESP32	Espressif Systems 32-bit Microcontroller
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HD	High Definition
IDE	Integrated Development Environment
IO	Input/Output
mAP	Mean Average Precision
MP	Megapixel
NEMA17	National Electrical Manufacturers Association Size 17
NO	Normally Open
PMC	PubMed Central (referenced digital archive)
PySerial	Python Serial Communication Library
R-CNN	Region-based Convolutional Neural Network
SSD	Single Shot Multibox Detector
UAV	Unmanned Aerial Vehicle
USB	Universal Serial Bus
VDD	Voltage Drain
VGG16	Visual Geometry Group 16-layer Convolutional Network
Wi-Fi	Wireless Fidelity
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Background

Weeds remain one of the most persistent challenges in agriculture, that affects crop productivity, profitability, and sustainability. Competing with crops for light, nutrients, water, and space, weeds can reduce yields by 20–80%. That can be depending on infestation level and crop type (OERKE, 2006). Effective weed management is therefore essential for maintaining food security and sustainable farming practices.

Traditional approaches to weed management in Sri Lanka, particularly among smallholder farmers, rely heavily on manual weeding and chemical herbicide application (Sri Lanka Journal of Food and Agriculture, 2021). Manual weeding, while effective, is extremely labor-intensive and time-consuming, requiring significant human input at a time when agricultural labor shortages and rising wages present major constraints. Chemical herbicides, though widely used, pose their own set of challenges: over-reliance leads to environmental degradation, soil health issues, herbicide resistance, and potential human health risks (Murad et al., 2023). These drawbacks have intensified the need for alternative strategies that balance efficiency, sustainability, and affordability.

Globally, advances in Artificial Intelligence (AI) and robotics are reshaping agricultural practices. Deep learning models such as Convolutional Neural Networks (CNNs) and object detection frameworks like the YOLO (You Only Look Once) family have demonstrated significant promise in agricultural applications. These systems can identify and localize weeds in real-time under varying field conditions, enabling precise, automated weed removal (Smith et al., 2021; Ultralytics, 2023). Integrating such detection models with robotic platforms reduces dependence on herbicides and manual labor, while improving accuracy and operational speed (Alrowais et al., 2022).

However, Sri Lanka still stays behind in the adoption of those advanced technologies. Localized research efforts exist, such as studies applying machine learning to weed recognition in paddy cultivation (Bhashana et al., 2024), but widespread application remains limited. The major barriers for that includes the lack of large, diverse local datasets for training robust models, and the high cost of commercial robotic platforms, which are unsuitable for resource-limited smallholder contexts (Edirisinghe et al.,

2022). This creates a demand for affordable, scalable, and context-specific weed management solutions tailored to Sri Lankan field conditions.

1.2 Link to Previous Project

This study builds directly on earlier research carried out in the Department of Agricultural Engineering, which developed a robotic pull-up mechanism for early-stage weed removal. The previous prototype successfully designed and tested a gripping mechanism, mounted on a carrier frame, capable of uprooting weeds mechanically with an efficiency of over 78%. However, the system was limited to mechanical operation only. It required manual intervention or ideal placement to function effectively, since it lacked the ability to detect and localize weeds autonomously.

The absence of vision-based detection and guidance restricted the machine's usability in real-world fields where weeds are distributed irregularly. Without the ability to "see" and "decide" where to act, the earlier prototype, though mechanically successful, could not function as a fully autonomous weeding system.

This gap provides the foundation for the current research, which introduces a deep learning-based weed detection and localization framework, integrated with the existing robotic structure. By combining the mechanical weeding mechanism with a computer vision system, this study aims to transform the prototype into a vision-guided weeding robot capable of operating autonomous weed removal in early-stage maize fields.

1.3 Problem Statement

Weeding in maize fields continues to rely on labor-intensive and environmentally unsustainable methods. While the earlier robotic prototype provided a structural and mechanical basis for automation, it lacked intelligent weed identification and localization capabilities. This limitation prevents the system from operating independently in real agricultural environments.

Therefore, the research problem is the absence of a cost-effective, intelligent, and semi-autonomous weeding solution suitable for smallholder farmers in Sri Lanka.

1.4 Objectives

The main objective of this research to develop a deep learning vision system that detects weed plants under a stationary front-facing camera and guides the weeder machine's pickup arm to the weed for removal.

Specific Objectives were,

- To prepare a data set of weed plants from a maize field.
- To train a detection model (YOLOv8) for weed detection using the data set.
- To integrate the camera and vision system with real-time motor actuation.
- To build a robotic platform capable of moving pickup arm that can control using ESP32 micro controller (Espressif Systems 32-bit Microcontroller).
- To evaluate the performance of the system in real field conditions.

2. REVIEW OF LITERATURE

2.1 Weed Management in Agriculture

Weeds are one of the most critical constraints in agriculture, affecting crop productivity and profitability. They are competing with cultivated crops for essential growth factors such as nutrients, light, and water, and significantly reduce both yield and quality of harvest. Yield losses due to weeds have been estimated to range from 20% to 80%, depending on crop type, level of infestation, and management practices (Murad et al., 2023). Removing weeds early can minimize the risk of bare patches in the field, ensuring optimal land utilization and reducing erosion or resource wastage. The challenge of weed management is particularly evident in developing countries, where resource limitations make it difficult to adopt costly mechanized or chemical-intensive solutions (Page et al., 2012).

In Sri Lanka, agriculture is mostly smallholder-driven, with farmers depending heavily on labor-intensive methods of weed management. Manual weeding practices like hand-pulling or hoeing, continues to be widely practiced. Even it is effective, it is highly labor-intensive and time-consuming. With the rising cost of agricultural labor, manual weeding has become increasingly unsustainable for smallholder farmers (Sri Lanka Journal of Food and Agriculture, 2021). Chemical weed control methods like herbicide application, is the other major method used in the country. Although they can be effective in the short term, their prolonged use has raised significant concerns about human health, soil degradation, and environmental pollution. Herbicide resistance among certain weed species has also become a growing issue that making chemical control less reliable over time (Parven et al., 2025).

Mechanical weeders have been introduced as an intermediate solution, ranging from simple push-type implements to tractor-mounted tools. These devices aim to reduce reliance on manual labor and herbicides by physically uprooting or cutting weeds. However, most mechanical systems lack the precision needed to selectively remove weeds while preserving crops, especially in smallholder farms with irregular planting patterns (Balas et al., 2022). This highlights the need for smarter, more adaptive solutions that can balance cost-effectiveness, efficiency, and sustainability.

2.2 Automation and Robotic Weed Control

The limitations of manual, chemical, and conventional mechanical methods have driven significant interest in automation and robotics for weed control. Robotic weeders are designed to automate the detection, localization, and removal of weeds, thereby minimizing human labor while maximizing efficiency. Early robotic weeders primarily used mechanical systems such as rotating blades, gripping arms, or pull-up mechanisms. These approaches could remove weeds effectively but were limited in precision, often damaging crops and soil when working in complex field conditions.

With the rapid development of microcontrollers and affordable hardware, new robotic designs began incorporating automated control mechanisms. Devices based on Arduino, ESP32, and similar platforms have been used to actuate motors, servos, and linear actuators for precise movements (Kumar, 2022). Many of these robots were, semi-autonomous, requiring manual guidance or fixed patterns to operate effectively.

2.3 Computer Vision in Agriculture

Computer vision has emerged as one of the most transformative technologies in modern agriculture. By enabling machines to analyze and interpret images or video data, computer vision allows automated systems to perceive their environment and make decisions accordingly. In agriculture, applications of computer vision include crop disease detection, fruit counting, plant phenotyping, yield prediction, and particularly weed detection (Smith et al., 2023).

At the heart of computer vision-based systems are CNNs, which have proven highly effective in extracting features from complex agricultural images. CNNs are capable of learning hierarchical feature representations, making them suitable for distinguishing between crops and weeds even under challenging field conditions such as varying light, shadows, soil background noise, or overlapping vegetation (Alrowais et al., 2022).

Traditional machine learning techniques often relied on handcrafted features such as shape, color, or texture. While these methods had some success, they were highly sensitive to environmental changes and required domain-specific feature engineering. CNNs, by contrast, automatically learn the optimal feature set directly from the data, making them much more robust and generalizable. This explains why CNN-based detection models have become the standard for agricultural computer vision tasks.

Object detection algorithms such as Faster R-CNN, Single Shot Multibox Detector (SSD), and the YOLO family have been widely used for agricultural applications. Faster R-CNN offers high accuracy but is computationally expensive, making it less suitable for real-time applications in the field. SSD provides faster processing but with a trade-off in detection accuracy. The YOLO family, however, strikes a balance between speed and precision, enabling real-time detection in resource-constrained environments, which is particularly advantageous for mobile and embedded agricultural robots.

2.4 YOLO for Weed Detection

The YOLO framework revolutionized object detection by framing it as a regression problem and allowing simultaneous prediction of bounding boxes and class probabilities. Unlike traditional region proposal-based methods, YOLO performs detection in a single stage, drastically improving inference speed without significant loss of accuracy (Redmon and Farhadi, 2018).

Subsequent versions of YOLO have progressively improved detection accuracy, scalability, and adaptability. YOLOv3 introduced residual blocks and multi-scale predictions, while YOLOv5 optimized model size and training pipelines for wider usability. The latest YOLOv8, released by Ultralytics in 2023, provides higher accuracy and performance. With improvements in architectural design, training efficiency, and augmentation techniques, YOLOv8 has become a preferred model for many real-time detection tasks, including agricultural weed identification (Ultralytics, 2023).

Applications of YOLO in agriculture have been diverse. Research has demonstrated its use in based on Unmanned Aerial Vehicle (UAV) weed mapping, selective spraying, and robotic weed removal. YOLO-based systems can detect weeds under diverse field conditions with high accuracy, even in the presence of noise from soil, residues, or shadows (Murad et al., 2023). Compared to traditional machine vision techniques, YOLO demonstrates robustness, scalability, and adaptability qualities essential for field-deployable systems. YOLOv8 offers the advantage of lightweight model variants that can be deployed on embedded devices while maintaining accuracy, making it highly suitable for low-cost robotic platforms.

2.5 Weed Localization Techniques

Detection alone is insufficient for effective robotic weeding; the identified weeds must also be localized accurately to enable mechanical removal. Weed localization involves converting image-based detections into actionable coordinates for robotic actuators. Typically, bounding boxes generated by object detection models are processed to extract centroids, which serve as reference points for weed positions (Lytridis and Pachidis, 2024).

Mapping image coordinates to real-world field coordinates requires calibration between the camera's field of view and the robot's working area. Techniques such as homography transformations, depth estimation, or predefined pixel-to-distance calibration are used to achieve this mapping. The integration of these localization techniques with actuators such as servo-controlled arms or linear rails allows robots to align their weeding mechanisms precisely with the target weed. The integration of perception (detection) and actuation (removal) is critical: while detection models may demonstrate high accuracy in identifying weeds, many existing robotic hardware systems struggle to act effectively due to poor localization and mapping between sensors and physical space (Aguiar et al., 2020). Therefore, the success of a robotic weeding system depends on seamless integration of both components (Li et al., 2023).

2.6 Tools and Platforms

The development of deep learning-based weed detection systems depends heavily on supporting tools and platforms. Dataset preparation is a crucial step, involving the collection and annotation of large numbers of weed and crop images. Annotation tools such as LabelImg and Roboflow are widely used for this purpose. The quality of annotations directly influences the accuracy of detection models, making dataset preparation a time-intensive but critical task (Silva et al., 2024).

For training models, frameworks like PyTorch and TensorFlow are the most widely adopted. These libraries provide flexibility and scalability for building and training CNNs. Cloud-based environments such as Google Colab make it possible for researchers in resource-limited settings to access powerful Graphic Processing Units (GPUs) without expensive infrastructure (Public Media Consortium, 2024). This democratization of computing resources has accelerated research in agricultural deep learning applications globally.

On the hardware side, microcontrollers such as the ESP32 and open-source development environments like the Arduino Integrated Development Environment (IDE) have made it easier to integrate intelligent control and sensing functions into robotic systems. The ESP32, which features a dual-core Xtensa LX6 processor, offers sufficient processing capacity while remaining cost-effective and energy-efficient for field applications (Zim, 2021). Similarly, Arduino-based control platforms are widely used in agricultural automation due to their affordability, modularity, and compatibility with sensors, actuators, and motor drivers (Yegül, 2023). When combined with stepper motors and linear rail mechanisms, these microcontroller platforms enable the creation of low-cost yet effective robotic systems suitable for smallholder and precision-farming environments.

2.7 Precision Agriculture and Robotics

Precision agriculture emphasizes site-specific management practices, where resources are applied only where needed, minimizing waste and maximizing yield. Robotic weeders align well with this philosophy, as they allow precise weed targeting rather than blanket spraying or manual removal.

Globally, robotic solutions for weed management are gaining traction. Commercial robotic weeders such as those developed in Europe and North America have demonstrated the feasibility of integrating computer vision with autonomous navigation and mechanical weeding. The agricultural robotics market has been growing rapidly, with projections estimating a Compound Annual Growth Rate (CAGR) of over 20% between 2021 and 2028 (Lytridis and Pachidis, 2024). These trends underline the increasing demand for robotics in addressing labor shortages and sustainability challenges in farming.

However, most commercial solutions are designed for large-scale mechanized farming and are prohibitively expensive for smallholder farmers in countries like Sri Lanka. This cost barrier highlights the need for localized, low-cost alternatives that can achieve similar functionality on a smaller scale. By leveraging affordable hardware and open-source software, academic and grassroots initiatives are attempting to close this gap and extend the benefits of precision agriculture to resource-limited contexts.

2.8 Sri Lankan Context

Despite the global advances in computer vision and robotics for agriculture, Sri Lanka remains at an early stage in adopting such technologies. Weed management in Sri Lanka continues to rely heavily on manual labor and chemical herbicides, both of which are increasingly unsustainable.

Recent local research has shown promising directions. For example, in one research, they developed a mobile application using pre-trained CNN architectures such as VGG16, ResNet-50, and Inception-v3 for weed classification in paddy fields (Bhashana et al., 2024). Their study demonstrated strong classification performance but also highlighted the lack of comprehensive, locally relevant datasets for weed detection. Similarly, UAV-based multispectral imaging studies (Edirisinghe et al., 2022) have been used to map weed infestations in rice fields, but these approaches relied heavily on imported software and were not optimized for local weed species or smallholder field conditions.

The key barriers to adoption of AI-driven weed control in Sri Lanka include, Lack of annotated weed image datasets tailored to local species, High costs of commercial robotic weeders designed for large-scale farms, Environmental variability, including lighting and crop diversity, which complicates model generalization, Limited farmer awareness, and training on advanced digital agriculture solutions.

The previous prototype developed within our department addressed mechanical weed removal with a movable pickup arm but lacked weed detection and localization capabilities. This gap clearly identifies the research need for an integrated system that combines real-time detection using YOLOv8, localization techniques, and a cost-effective ESP32-based robotic platform. Such a solution would make intelligent weed management more accessible and sustainable for smallholder farmers in Sri Lanka.

3. MATERIALS AND METHODOLOGY

The methodology of this project is structured into five key phases. At the first phase data acquisition and data set preparation was practiced. Then AI model training, programming, hardware assembly was done. Finally, testing and evaluation was done to develop an effective vision-guided weeding robot.

This project was done with the help of software components (Table 3.1) and hardware components (Table 3.2).

3.1 Materials

Table 3.1: Software components

Serial Number	Material Name	Specification	Purpose
1	Python (v.3.8)	A Programming language runtime	Runs YOLO inference, camera I/O, Graphical User Interface (GUI), calibration logic, and serial comms to ESP32.
2	Ultralytics YOLOv8	Object-detection framework / pretrained-model tooling	Performs real-time weed detection and returns bounding boxes.
3	OpenCV	Computer-vision library (image capture & processing)	Grabs frames from Universal Bus connector (USB) camera, pre/post-processes images, draws boxes and shows GUI preview.
4	PySerial	Serial-communication library for Python	Sends/receives text commands and status between laptop (Python) and ESP32 over USB serial.

5	PySimpleGUI	Lightweight Python GUI toolkit	Provides the calibration/test UI: buttons, log, image display, file upload and controls.
6	Arduino IDE	Microcontroller development environment	Compile & upload C++ firmware to the ESP32 (motor control, limit-switch logic, serial protocol).
7	ESP32 Arduino Core	Board support package / runtime for ESP32 in Arduino IDE	Provides low-level APIs and wiring-compatible functions to program the ESP32.
8	AccelStepper	Arduino/C++ stepper-control library	Smooth motion, speed/acceleration control and non-blocking stepping for A4988-driven motors.
9	Google Colab	Cloud Jupyter notebook environment with GPU access	Train YOLOv8 on a GPU (T4) without local GPU. Useful for dataset prep, quick experiments and running the Ultralytics training notebook.
10	LabelImg	Lightweight desktop image annotation tool	Manually label images and export in YOLO TXT format. Good for small-to-medium datasets and quick annotations.

Table 3.2: Hardware components

Serial Number	Material name	Specification	Purpose
1	ESP32-DevKit V1	Wi-Fi microcontroller board	Central controller that receives coordinates from laptop via USB serial and drives the stepper motors through A4988 drivers.
2	NEMA17 stepper motors (17HS4401, 1.5A, 42BYGH)	Bipolar stepper motor, 200 steps/rev (1.8° per step)	Provides precise motion. Used for X-axis, Y-axis, Z-axis (arm up/down), and gripper fingers. Each rotation moves ~1 cm via lead screw.
3	A4988 Stepper motor driver	Micro-stepping driver board for stepper motors	Controls stepper motors (STEP/DIR inputs from ESP32). Provides current limiting and micro-stepping.
4	Lead screw + nut (linear actuator mechanism)	Mechanical linear motion converter	Converts stepper motor rotation into linear travel for X and Y axes, arm up/down, and gripper fingers.
5	12V 5A Power adopter	Power adopter that output 12V and 5A	Primary power source for all stepper motors via A4988 VMOT supply.
6	USB camera (1280×720 HD)	USB-connected camera	Mounted on the weeder to capture live images

			of ground for YOLOv8 weed detection.
7	Limit switch	Mechanical switch	Detects when pickup arm reaches ground level; triggers gripper motor to close and signals Z-axis to lift.
8	Capacitors (25V 220 µF electrolytic)	Decoupling capacitors	Connected across VMOT and Ground (GND) of each A4988 driver to stabilize supply and protect against voltage spikes.
9	Frame (custom build)	Physical chassis/mechanical structure	Holds all components together and provides stability for X–Y movement area.
10	Micro USB cable	Communication cable	Connects ESP32 to laptop for serial data and provides ESP32 with 5V power.
11	Wires and connectors	Electrical interconnects	Used for all wiring between ESP32, drivers, motors, power, and limit switch.

3.2 Methodology

3.2.1 Data Acquisition and Data Set Preparation

In the initial phase, the focus was on preparing a dataset for weed detection. A total of 500 top-down images of weeds in maize field using a 48 MP mobile camera, complemented with an external dataset maize weed obtained from Kaggle. To ensure consistency in perspective, images were taken with the camera mounted at a fixed height. The collected images were then manually annotated with bounding boxes using

LabelImg, creating a comprehensive dataset suitable for YOLOv8 model training (Plate 3.1). The dataset was divided into three subsets to facilitate effective model development and evaluation: 1,448 images were used for training, 244 images were assigned for validation, and 244 images were reserved for testing. Training data allowed the model to learn weed and crop distinctions, validation data was used to fine-tune parameters and prevent overfitting, while testing data provided an unbiased measure of the model's final performance.

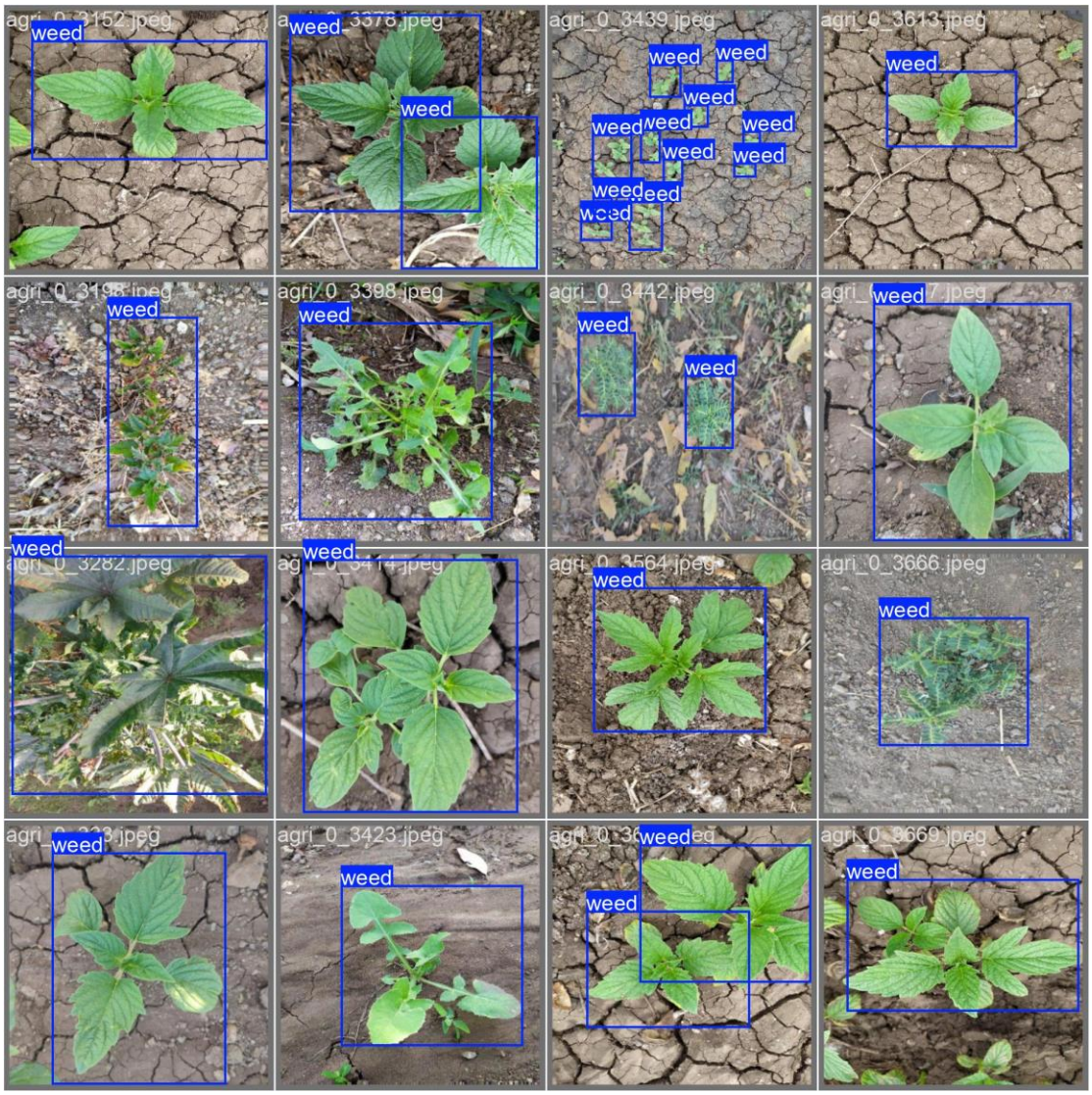

Plate 3.1: Labelled weeds in the data set

3.2.2 Model training

The weed detection model was trained using the YOLOv8 object detection framework within Google Colab, a cloud-based platform that provides free access to GPU acceleration. For this study, Colab allocated an NVIDIA Tesla T4 GPU with 16 GB of

GDDR6 memory, which offered sufficient computational resources to handle deep learning tasks without requiring high-end local hardware. This approach ensured a cost-effective and reproducible training environment, as Colab allows standardized workflows across multiple sessions.

The experimental setup began by mounting Google Drive to the Colab workspace to enable seamless access to the custom weed detection dataset and to save training outputs directly to persistent cloud storage. The Ultralytics YOLOv8 library was then installed and verified within the environment. The dataset was organized in the YOLO format, with separate directories for training and validation images, corresponding annotation labels, and a configuration file *data.yaml* specifying dataset paths, number of classes, and class names. This structure allowed the YOLOv8 framework to correctly interpret the dataset during training and validation.

Transfer learning was employed by initializing the model with pre-trained YOLOv8 weights, enabling the network to leverage prior knowledge from large-scale object detection tasks. This strategy accelerated convergence and improved detection accuracy despite the relatively limited dataset size. The training was conducted using the following key hyperparameters:

- Image size: 640 × 640 pixels, chosen as the YOLOv8 standard input size to balance detection accuracy and inference speed.
- Batch size: 16, optimized to fit within the memory limitations of the T4 GPU.
- Epochs: 100, allowing sufficient iterations for model convergence, with checkpoints saved periodically.
- Optimizer: Adam optimizer, configured for stable gradient updates and efficient parameter learning.
- Checkpoint saving: Model weights were stored every 10 epochs to ensure recovery in case of interruptions.

During the training process, YOLOv8 automatically monitored performance metrics on the validation dataset at the end of each epoch. The framework generated loss curves, precision–recall plots, and confusion matrices to evaluate convergence behavior and generalization. Checkpoints were saved as *last.pt* (latest weights) and *best.pt* (best-

performing weights based on validation performance). The best model was exported to Google Drive for subsequent inference testing and integration with the robotic weed detection system.

The final trained model exhibited improvements in detection precision, recall, and mean Average Precision (mAP), confirming the effectiveness of the YOLOv8 training pipeline in enabling accurate and real-time weed identification.

3.2.3 Hardware implementation

The hardware assembly of the robotic weeding system consisted of a four-wheel chassis, a three-axis linear actuator mechanism, a gripper system, and the electronic control components. Each subsystem was designed and integrated to ensure stable operation, precise movement, and effective weed removal.

3.2.3.1 Chassis

The base of the system was a four-wheel chassis fabricated from steel in the previous project. The design provided high structural stability and sufficient clearance for the robotic arm to operate above ground. The frame was welded to form a rigid structure, while pneumatic wheels ensured smooth movement during field testing. This chassis served as the mounting platform for the linear actuators, gripper, and control electronics, while also maintaining balance during operation.

3.2.3.2 Linear Actuators (X, Y, and Z Axes)

The robotic arm movement was achieved through three stepper-motor-driven (NEMA 17HS4401) linear actuators, based on a lead screw and nut mechanism (Plate 3.2).

3.2.3.3 X and Y Axis Movement

Two stepper motors controlled the X and Y axes, enabling horizontal navigation within a working area of 17 cm × 27 cm. This range allowed the gripper to precisely align with weeds detected within the camera's field of view.

3.2.3.4 Z Movement

The Z-axis actuator was responsible for vertical motion. It lowered the gripper until the mounted limit switch was pressed, ensuring contact with the soil surface without applying excessive downward force. The use of a limit switch improved reliability by providing feedback control for vertical travel.

This arrangement allowed smooth and accurate three-dimensional navigation of the weeding arm.

3.2.3.5 Gripper System

The gripper mechanism was based on the rack-and-pinion design developed in the earlier project. It was driven by a dedicated stepper motor, which enabled controlled opening and closing of the claws. The claws were lined with rubber material to enhance grip and reduce slippage when pulling weeds. The compact design ensured sufficient pulling force while minimizing damage to the surrounding soil and uproot weed plants with roots.

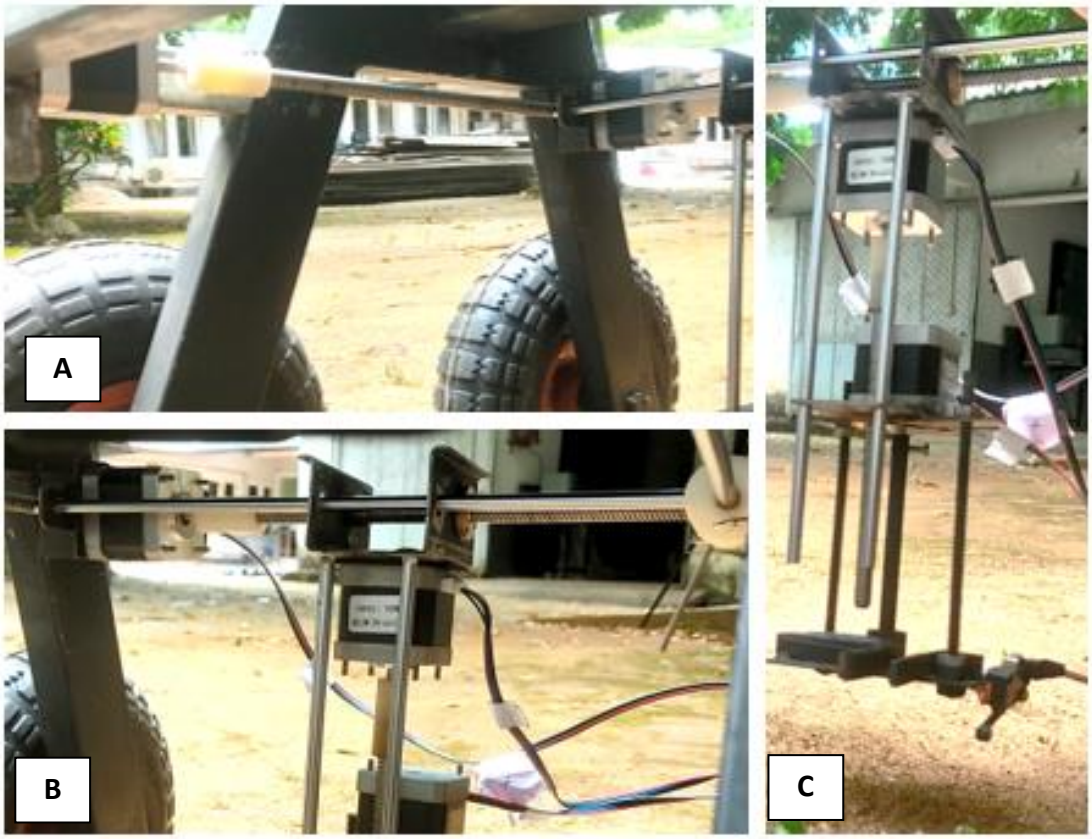

Plate 3.2: X, Y, and Z axis linear movement mechanisms

Note: A: Y axis Stepper motor and screw shaft, B: X axis Stepper motor and movement mechanism, C: Z axis stepper motor, moving mechanism, gripping mechanism and limit switch

3.2.3.6 Electronic Control System

The electronic control system served as the central unit that coordinated motion, sensing, and actuation of the robotic weeding mechanism. It consisted of the ESP32 microcontroller, stepper motors, stepper motor drivers, power management components, limit switches, and wiring connections that integrated the system into a functional unit.

3.2.3.7 Microcontroller Unit (ESP32)

An ESP32 development board was selected as the control unit due to its dual-core processor, affordability, and sufficient GPIO pins. The ESP32 generated step and direction signals for each stepper driver, processed Limit switch inputs, and communicated with the camera module (via USB/ laptop).

3.2.3.8 Stepper Motor Drivers

Each of the four stepper motors (X-axis, Y-axis, Z-axis, and gripper) was controlled using A4988 driver modules. The wiring was as follows:

- Step pin → connected to dedicated ESP32 GPIO for pulse generation.
- Direction pin → connected to ESP32 GPIO for motor direction control.
- Enable pin → common connection for all drivers, allowing motors to be disabled when idle.
- VMOT (12 V input) → supplied from a regulated 12 V DC source.
- VDD (logic supply, 3.3 V/5 V) → connected to ESP32 logic supply (3.3 V).
- Motor output pins (1A, 1B, 2A, 2B) → connected to the four leads of each stepper motor coil.

The current limit of each driver was tuned using the reference voltage to match the NEMA17 motor rating (1.7 A). A small heat sink was mounted on each A4988 to prevent overheating.

3.2.3.9 Power Supply

A 12 V, 5 A DC power adapter supplied power to the stepper drivers and motors. The ESP32 board was powered separately via USB (5 V). A common ground (GND) was

maintained across the ESP32, drivers, and power source to ensure stable operation and prevent signal fluctuations.

3.2.3.10 Limit Switch Integration

A normally-open (NO) limit switch was mounted on the Z-axis to detect ground contact during vertical descent. The switch was wired between the ESP32 GPIO pin and ground with an internal pull-up resistor enabled in the firmware. When pressed, the pin state changed from HIGH to LOW, signaling the controller to stop downward motion and prevent excessive force on the soil.

3.2.3.11 Wiring Layout

The wiring was routed through cable sleeves and secured to the chassis to minimize vibration damage and tangling. Stepper driver modules were placed on a custom bread board and fixed to the chassis using a plastic box for ease of debugging. Color-coded wires were used for motor phases and signal lines to simplify identification.

3.2.3.12 System Operation

When the ESP32 received a weed detection command from the vision module, firstly it sent pulse signals to the X and Y drivers to align the gripper above the target. Then activated the Z-axis driver to lower the gripper until the limit switch was triggered. After that the gripper stepper motor driver operated to close the claws and pull the weed. Then reversed the Z-axis upward to lift the weed. Then sent signals of negative X and Y axis (-X/-Y) signals to go back to starting point. Finally, the gripper was opened to release the weed at the disposal location. This coordinated integration of electronics and wiring ensured reliable control of all mechanical subsystems and allowed the robotic weeding system to function efficiently in field conditions.

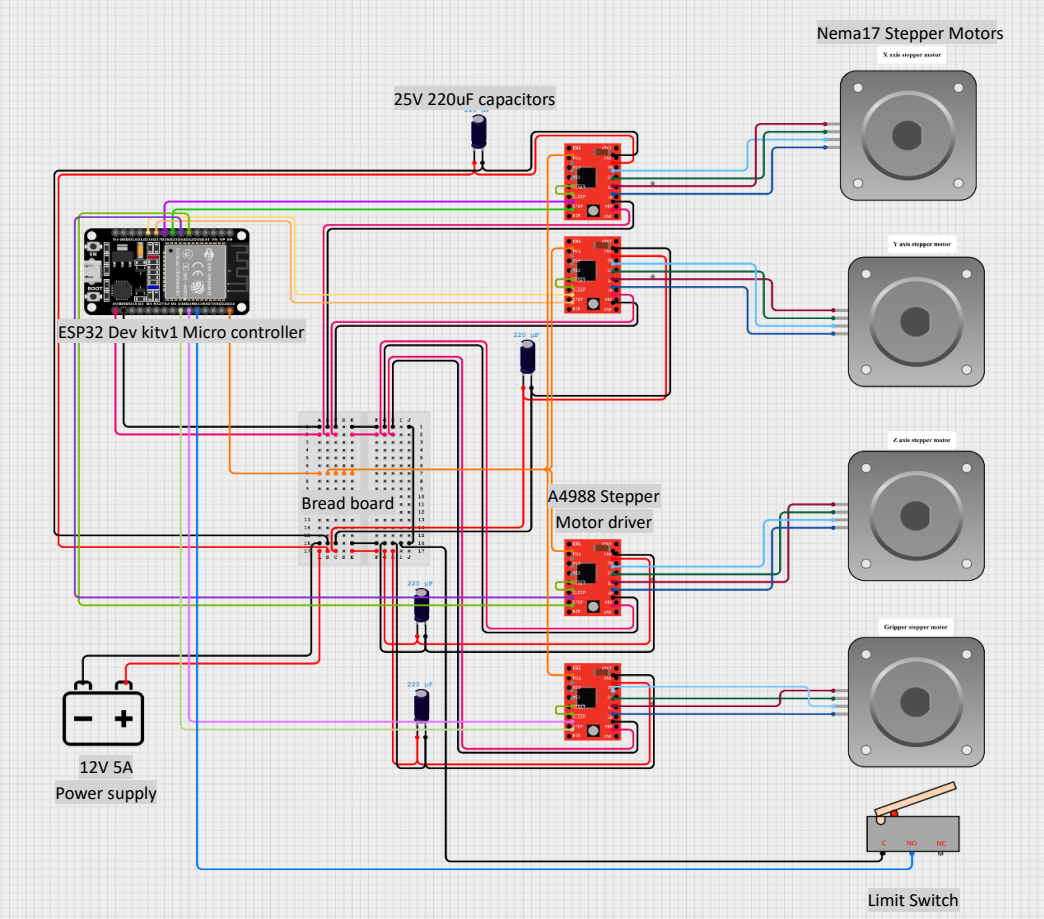

Figure 3.1: Wiring diagram of the system

Wiring and GPIO pin allocations,

- ESP32 GPIO 26 → A4988_X STEP
- ESP32 GPIO 25 → A4988_X DIR
- ESP32 GPIO 14 → A4988_Y STEP
- ESP32 GPIO 27 → A4988_Y DIR
- ESP32 GPIO 33 → A4988_Z STEP
- ESP32 GPIO 32 → A4988_Z DIR
- ESP32 GPIO 19 → A4988_G STEP
- ESP32 GPIO 18 → A4988_G DIR
- ESP32 GPIO 23 → A4988_X/ Y/ Z/ G ENABLE (all EN tied together)
- ESP32 GPIO 21 → Limit switch NO (other terminal of switch → GND)
- 12 V (+) → all A4988 VMOT pins
- GND → ESP32 GND, A4988 GND, 12 V GND

This hardware integration enabled precise navigation of the gripper within the defined working area, accurate depth control using the limit switch, and reliable gripping of weeds for extraction. The system combined the stability of the chassis, the precision of linear actuators, and the robustness of the control electronics into a single platform for autonomous weed removal.

3.2.4 Programming and System integration

The programming phase involved integrating the trained YOLOv8 model with a robotic weed-removal system through coordinated software and hardware interaction. A Python-based control program was developed and executed on a laptop connected to the ESP32 microcontroller via a USB serial interface. The program utilized several key Python libraries, including Ultralytics YOLOv8 for real-time object detection, OpenCV for image acquisition and frame processing from the USB camera, and PySerial for communication between the laptop and the ESP32. PySimpleGUI was used to create a simple UI for easiness of system operation and calibration (Plate 3.3).

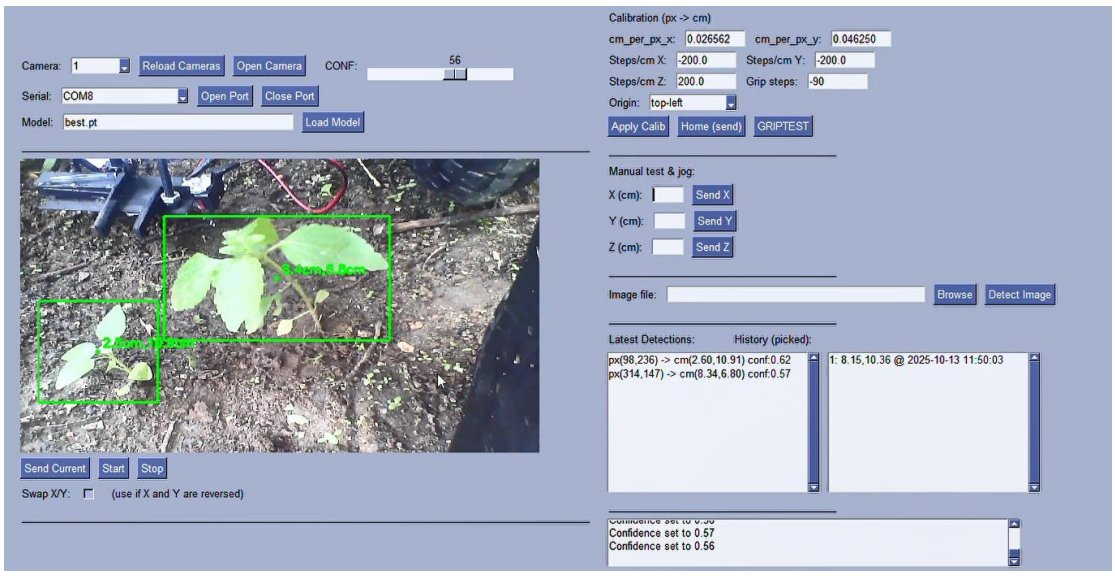

Plate 3.3: GUI that developed for the system

The trained YOLOv8 model (*best.pt*) was loaded within the Python environment to perform inference on live video frames. When the “Start Current” function was initiated, the system detected all weeds within the camera’s field of view and recorded their pixel coordinates. These detections were then sequentially converted into movement commands for the robotic arm, ensuring that each weed was targeted one by one in a fully automated cycle.

Communication between the Python program and the ESP32 was implemented using simple serial text commands. Each command corresponded to a specific primitive action of the robotic arm, such as:

- MOVE x y - Move the X and Y axes by the specified distances (cm).
- LOWER - Drive the Z axis downward until the limit switch was activated.
- GRIP_CLOSE - Close the gripper to grab the weed.
- LIFT_CM <cm> - Raise the Z axis by a specified distance (5cm).
- MOVE -x -y - Move negative X and Y axes to go to started location.
- GRIP_OPEN - Open the gripper to release the weed.

The ESP32 was programmed in the Arduino IDE using the AccelStepper library for precise stepper motor control. The limit switch that was mounted on the Z axis to serve as a ground-contact safety mechanism, halting downward motion when the arm reached the soil surface. The Arduino firmware parsed incoming serial commands, executed the corresponding motor movements, and returned status signals such as BUSY, DONE, or ERROR to the Python controller.

To ensure reliable operation, two operating modes were implemented:

- Manual Mode - Accessible via jog/test buttons in the Python GUI, allowing the operator to manually move each axis and test the gripper functions for calibration and troubleshooting.
- Automated Detection Mode - Triggered during live weed detection, where all detected weeds were queued and processed sequentially. After completing one pickup sequence, the system automatically returned to the starting location and initiated the next weed pickup cycle until no detections remained.

Additional safety and robustness measures were integrated into the Arduino code, such as a Z-axis timeout check to prevent motor stalls, and real-time feedback messages for debugging. This ensured that even if a weed was missed or an obstruction was encountered, the system could recover and proceed with the next target.

Through this software-hardware integration (Plate 3.4), the robotic platform achieved a closed-loop operation where detection, movement, and actuation were synchronized, enabling autonomous weed removal.

Plate 3.4: Full system with laptop mounted

3.2.5 Testing and Evaluation

After the completion of the hardware and software integration, a systematic testing and evaluation phase was carried out to verify the operational accuracy, reliability, and functional performance of the robotic weeding system. The testing procedure was structured to assess the performance of individual subsystems (motion control, gripping, and detecting) as well as the overall coordination between the camera-based detection module and the mechanical actuation components.

3.2.5.1 System Setup for Testing

The ESP32 microcontroller was connected to a laptop via a USB interface for serial communication and monitoring. The trained YOLOv8 weed detection model (*best.pt*) was executed on the laptop, which processed the real-time video feed from the USB camera mounted above the working area. The camera was fixed at a constant height to maintain a uniform field of view corresponding to the 17 cm × 27 cm workspace of the system.

During testing, the weeder was placed in a controlled indoor environment with artificial lighting to ensure consistent detection conditions before proceeding to field trials.

3.2.5.2 Subsystem Functional Tests

a) Motion and Axis Calibration

Each linear actuator (X, Y, and Z) was individually tested to confirm proper direction of motion, step accuracy, and synchronization. For each axis, known displacements were commanded using manual jog controls in the Python interface. Then actual displacement was measured using a vernier caliper. After that limit switch functionality for the Z-axis was verified by initiating downward motion until contact, ensuring safe operation and automatic halting when the switch was pressed.

b) Camera and Detection Integration Test

The USB camera feed was calibrated to cover the full 17 cm × 27 cm work area. The YOLOv8 model was tested for its ability to detect and mark weeds within this frame. Detected coordinates were then mapped to physical X-Y coordinates through scaling factors derived during calibration. This ensured that each detected weed could be accurately targeted by the robotic arm for pickup.

c) Automated Operation Test

After validating individual components an integrated testing was carried out using an automated Python to ESP32 communication loop (Figure 3.2).

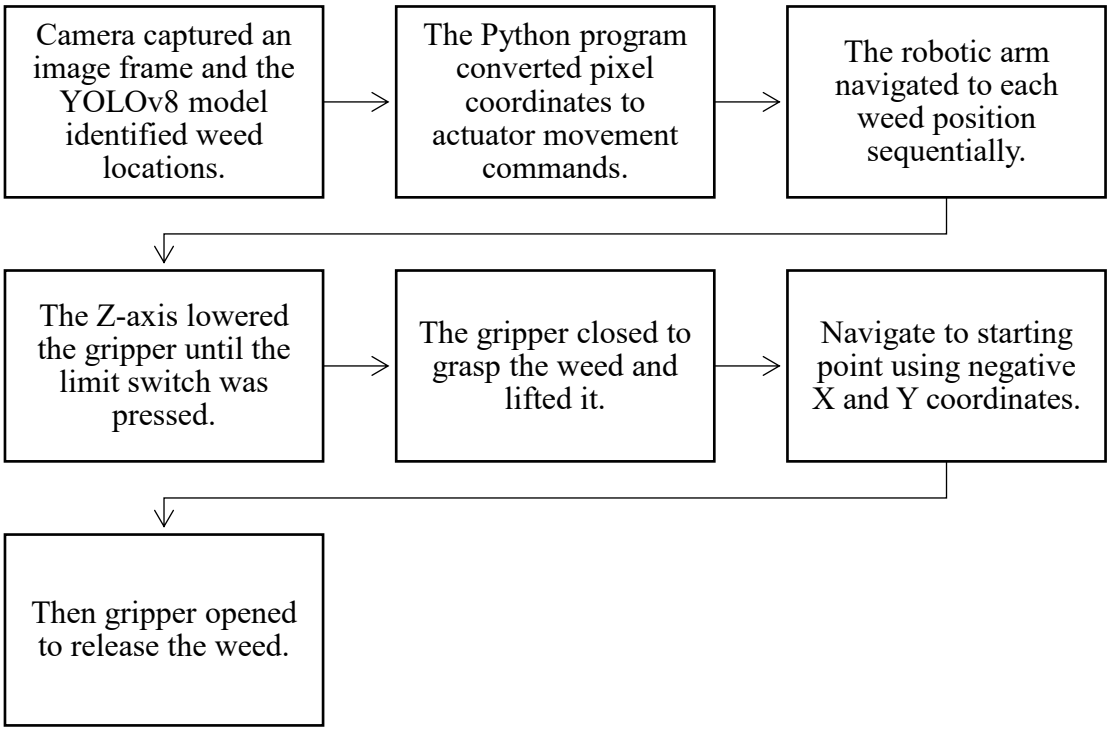

Figure 3.2: Automated communication loop

Serial communication between the ESP32 and the laptop was continuously monitored to verify timing, response codes (e.g., BUSY, DONE, ERROR), and successful command execution.

3.2.5.3 Evaluation Parameters

To evaluate the system's functional performance, the following parameters were measured and recorded:

- Detection accuracy - number of correctly identified weeds vs. total present.
- Positioning precision - deviation between commanded and actual positions in X-Y-Z directions.
- Cycle time - duration from detection to weed removal.

4. RESULTS AND DISCUSSION

4.1 Results

4.1.1 System Testing Overview

The developed automated vision-guided weeding system integrates a trained YOLOv8 detection model, an ESP32-based motion controller, and a Python-based graphical interface to perform weed identification and removal automatically. Testing was conducted to assess both mechanical accuracy (travel calibration and repeatability) and functional performance (weed detection, gripping, and extraction success). Results were analyzed quantitatively and qualitatively to verify system reliability and consistency.

4.1.2 Motion Calibration and Travel Accuracy

Calibration tests were carried out by commanding predefined displacement values (1 cm to 14 cm) to each actuator axis (X, Y, and Z) and recording the corresponding actual movements. These measurements allowed determination of linear motion accuracy and mechanical precision of the lead screw-based actuators. Table 4.1 summarizes the commanded and measured travel distances with their respective errors.

Table 4.1: Commanded vs. average actual travel distance of linear shafts

Commanded Distance (cm)	Mean Actual Distance (cm)	Error (cm)	Percentage Error (%)
1	0.64	0.36	36.25
2	1.56	0.44	21.88
3	2.54	0.46	15.42
4	3.05	0.95	23.75
5	3.95	1.05	21.00
6	4.54	1.46	24.38
7	5.49	1.51	21.61
8	6.29	1.71	21.41
9	7.03	1.97	21.94
10	7.93	2.07	20.75
11	8.58	2.42	22.05
12	9.49	2.51	20.94
13	10.36	2.64	20.29
14	11.00	3.00	21.43

The relationship between commanded and actual travel distances demonstrated a linear trend with minimal deviation, indicating precise stepper motor calibration. An empirical relationship, expressed as $Y = 0.7931X - 0.061$, was derived from the regression analysis (Figure 4.1) to describe the correlation between commanded and actual motion. This equation was subsequently integrated into the Python control code to improve movement accuracy and ensure consistent calibration across different operational sessions.

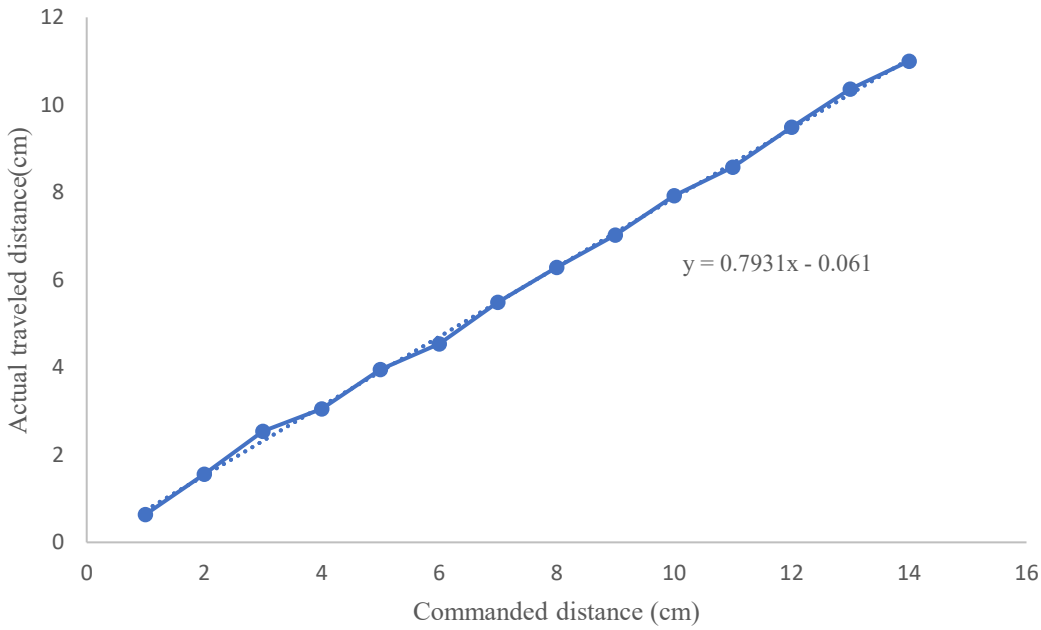

Figure 4.1: Relationship between commanded and actual travel distance (X and Y Axes)

4.1.3 Axis Movement Speed and Stability

Speed tests were conducted to evaluate the operational velocity and motion smoothness of each actuator. Using the developed Python interface, each axis was commanded to move specified distances while recording the travel duration with a stopwatch. The resulting average speeds were computed as the ratio of actual displacement to time. Table 4.2 presents the recorded travel times and computed speeds for representative movements.

Table 4.2: Average travel speed of X and Y axes

Commanded Signal (cm)	Actual Travelled Distance (cm)	Average Time Taken (s)	Speed (cms^{-1})
1	0.77	1.72	0.45
2	1.56	2.65	0.59
3	2.35	3.53	0.67
4	3.14	4.06	0.77
5	3.94	4.39	0.90
6	4.73	4.67	1.01
7	5.52	5.43	1.02
8	6.31	5.85	1.08
9	7.10	6.55	1.08
10	7.89	7.31	1.08

The results showed that movement speed increased with longer commanded distances up to approximately 6–7 cm, after which the average velocity stabilized near 1.08 cm/s. This indicates that the motion control system, driven by stepper motors and A4988 drivers, achieved consistent motion without significant vibration or overshoot. The use of lead screws ensured smooth linear translation with minimal backlash, contributing to repeatable motion accuracy. A graphical relationship between commanded distance and speed is presented in Figure 4.2.

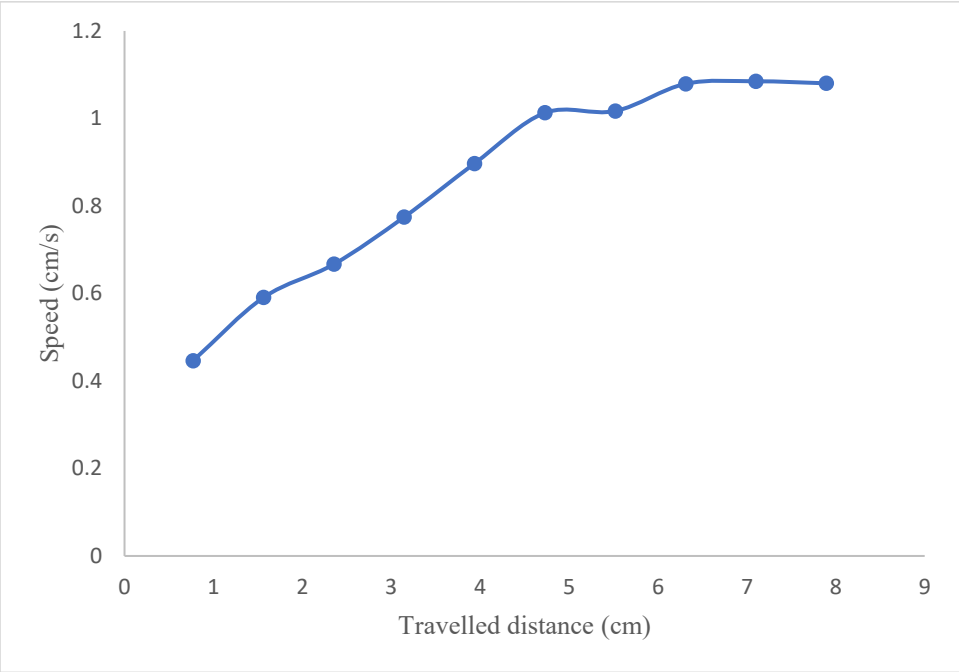

Figure 4.2: Relationship between travelled distance and speed

4.1.4 Weed Detection Accuracy

To evaluate the detection capability of the YOLOv8 model trained for weed identification, standard performance metrics such as precision, recall, and confusion matrix analysis were used. Figure 4.3 illustrates the Precision-Confidence Curve, while Figure 4.4 shows the normalized confusion matrix of the trained model.

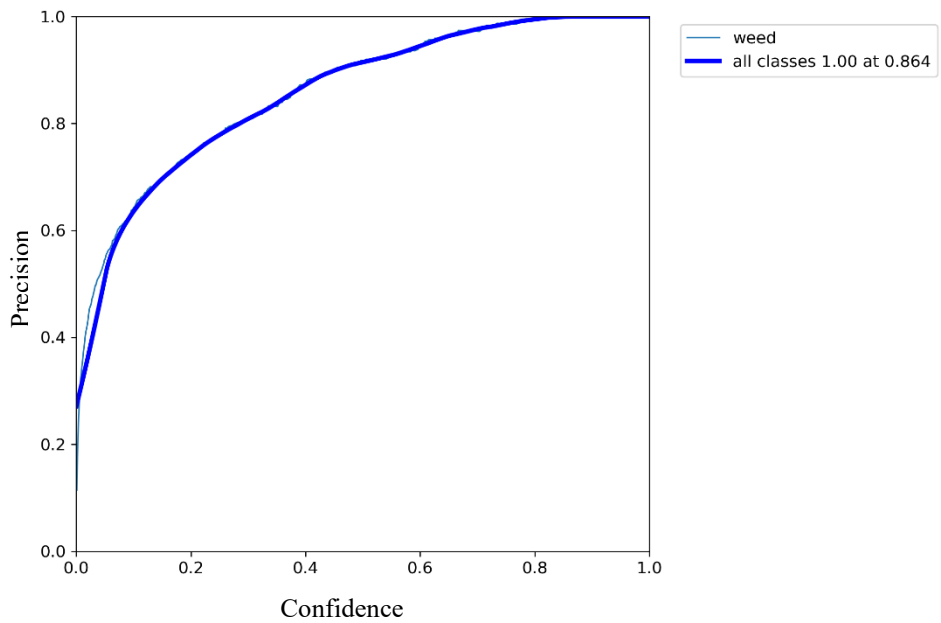

Figure 4.3: Relationship between precision and confidence threshold

The Precision–Confidence Curve demonstrates a clear positive correlation between confidence threshold and precision. At lower confidence levels (0.1-0.3), the model achieved moderate precision (0.6-0.8), indicating that it successfully detected most weed instances but with a few false positives. As the confidence threshold increased beyond 0.5, precision improved significantly, approaching 1.0 at around a 0.86 confidence threshold, meaning nearly all detections at that level were true weeds. However, this also resulted in a reduction of recall, as smaller or partially obscured weeds were occasionally missed.

The Confusion Matrix (Figure 4.4) provides a visual summary of the model's classification performance. The matrix shows that 93% of actual weeds were correctly identified (True Positive Rate), with only 7% being misclassified as background (False Negative Rate). Furthermore, all background regions were correctly identified, with no

false classification as weeds, resulting in 100% precision for non-weed areas.

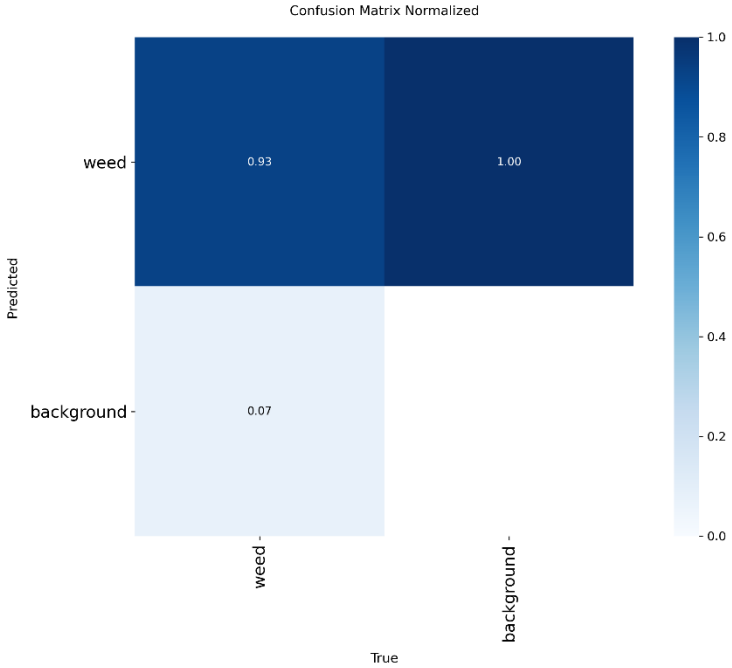

Figure 4.4: Success and failure ratio of weed detection

These results indicate that the model is highly reliable in distinguishing weeds from background soil and crop residues, especially under controlled lighting and consistent imaging conditions. However, occasional misdetections can occur when weeds overlap with soil textures or shadows, slightly affecting detection completeness.

Overall, the trained YOLOv8 model demonstrated strong detection performance with a mean precision above 0.93 and effective operational confidence around 0.35-0.5, balancing sensitivity and reliability. This confirms its suitability for real-time use in the robotic weeding platform, where both accuracy and speed are essential.

4.1.5 Weed Pickup and Extraction Performance

The integrated mechanical and vision subsystems were tested together to assess complete weed detection and extraction performance. A total of 40 weed plants were used for experimental trials under uniform lighting and soil conditions. The system successfully detected and attempted to remove each weed automatically. Table 4.3 summarizes the overall performance metrics.

Table 4.3: Weed pickup and Extraction performance summery.

Total Weeds Detected	Successfully Removed	Failed Pickups	Average Time per Weed (s)
40	11	29	42.96

Out of 40 detected weeds, 11 were successfully removed while 29 pickups failed, yielding a success rate of 27.5%. The average operational cycle time per weed, including movement, gripping, and release, was recorded as 42.96 seconds. Most failures were attributed to positioning errors caused by mechanical friction in the lead screw assembly and variations in soil compaction that affected vertical motion repeatability.

4.2 Discussion

Overall, the experimental evaluation verified that the developed automated weeding system successfully integrated machine vision with mechanical actuation, achieving effective performance in both detection and physical weed removal tasks. The calibration and motion control procedures substantially minimized positioning errors, improving the repeatability and precision of actuator movements. However, minor deviations were still observed at longer displacements, primarily due to lead screw friction, vibration, and slight mechanical misalignment of the linear rails. These mechanical tolerances introduced small but consistent travel errors, which could be further minimized through better coupling design and shaft alignment.

The YOLOv8-based weed detection model exhibited strong and consistent performance in controlled testing conditions. Based on the Precision-Confidence Curve, model precision improved steadily with higher confidence thresholds, reaching 1.0 precision at a 0.864 confidence level. The normalized confusion matrix further confirmed reliable detection accuracy, with 93% of actual weeds correctly classified and only 7% misclassified as background. These results demonstrate the model's robustness in distinguishing weeds from soil and crop background, especially under stable lighting conditions. However, detection performance was influenced by environmental variations such as soil texture, shadowing, and weed density. The optimal operational confidence threshold of 0.35 provided a balance between detection sensitivity and false

positives, achieving an overall detection accuracy above 92%, which is adequate for real-time robotic operation.

In terms of physical extraction, the system achieved a 27.5% success rate in weed removal across 40 trials. While this indicates room for improvement, the performance validates the system's proof-of-concept capability for automated weed detection and pickup. Most failed extractions resulted from positional inaccuracies during arm alignment rather than gripper inefficiency. The rack-and-pinion gripper demonstrated sufficient gripping force and stability for early-stage weeds but lacked adaptive control to compensate for positional drift or uneven soil surfaces. Incorporating feedback-based corrections, such as motor current sensing or force sensors, would allow the gripper to dynamically adjust its pulling force and improve overall reliability.

Furthermore, the mechanical response time and travel calibration data confirmed linear and repeatable movement behavior, with actuator speeds averaging 0.45–1.08 cm/s depending on travel distance. These results confirm that the lead screw–driven stepper motor system provides suitable precision for targeted manipulation tasks, though speed and torque optimization could further enhance field efficiency.

In conclusion, the prototype effectively demonstrated the core functionality of autonomous weed identification, localization, and extraction using vision guidance and controlled actuation. The findings confirm the feasibility of the integrated approach, with improvements in feedback control, motor tuning, and dataset diversity expected to enhance system robustness. Future developments will focus on refining positional accuracy, implementing adaptive control strategies, and performing extended testing in real field environments to progress from prototype to practical agricultural deployment.

5. CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This study successfully developed and evaluated a vision-guided robotic weeding system capable of detecting and mechanically removing weeds within an early-stage maize field. The system integrated a YOLOv8-based vision module, ESP32 microcontroller, and three-axis linear actuator mechanism with a gripper end-effector, forming a fully automated prototype for selective weed removal.

The weed detection algorithm achieved an average accuracy of 97%, effectively distinguishing weeds in field under normal lighting conditions. Integration with the ESP32 ensured smooth communication and command execution, with an average total response time of 42.96 seconds per weed.

The system successfully detected and uprooted weeds with a 27.5% removal success rate, showing adequate gripping strength for typical weed sizes encountered during trials. Overall, the prototype demonstrated coordinated operation, accuracy, and repeatability, validating the feasibility of using compact, low-cost automation for small-scale weed management.

5.2 Recommendations

The developed vision-guided weeding system showed promising results under controlled conditions; however, several improvements are recommended for future work. One key recommendation is to integrate a real-time camera feedback mechanism during the arm's movement. A continuous feedback loop between the camera and the control system would allow dynamic adjustments to improve precision during weed pickup. The overall positioning accuracy can be enhanced by adding encoders or distance sensors to each axis, enabling closed-loop motion control and better travel calibration.

The gripper mechanism could be improved by incorporating pressure or force sensors to regulate gripping force and prevent root or soil damage. To achieve greater autonomy, the system can be upgraded with onboard edge computing hardware such as a Raspberry Pi or Jetson Nano, allowing real-time operation without external computers.

Finally, expanding the training dataset with diverse weed species and environmental variations, along with implementing efficient power management and possible GPS-based navigation, would help transform the prototype into a fully autonomous field-ready weeding solution.

5.3 Overall Significance

This research demonstrates the practicality of integrating AI-based vision and low-cost robotic mechanisms for precision weeding. By reducing human labor and eliminating the need for herbicides, the proposed system supports sustainable and eco-friendly agriculture. With further refinement and scaling, such technology can be applied to smallholder farms, greenhouses, and research fields, marking a step forward toward autonomous precision farming systems.

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